

## Technical Datasheet for NTC Sensor with RS485 (Modbus RTU)

### 1. Product Overview

**WIN-SN-NTC-M** is a versatile and reliable solution for anyone needing **precise temperature monitoring** with robust communication capabilities. It measures temperature in various environments, from industrial facilities to greenhouses and smart homes. This board stands out because it can communicate its readings through an isolated RS485 interface. This type of communication is robust, making it ideal for long-distance data transmission and resistant to electrical interference, ensuring accurate and reliable measurements.

One of the key advantages of the WIN-SN-NTC-M-D is its flexibility in sensors compatibility.



This manual provides instructions for installation and use of the sensor board to measure temperature in the environment. The sensor board offers data output in Modbus RTU (RS485) and can be powered by DC power supply.

## 2. Sensor Specifications

- **Sensor Type:** NTC Thermistor (Model: MF52C1)
- **Sensor Specs:** 10KΩ, B-value 3950K ±1%
- **Probe Material:** SUS316 Stainless Steel with PVC cable
- **Operating Temperature:** -40°C to +105°C
- **Product dimension:** 85\*85\*40 mm
- **Certifications:**


## 3. Power Supply Connection

### DC Connection

- Use 12-24V@ 1A, Power supply.
- Don't make loose connection.

## 4. Configuration Settings

- Communication Speed 9600 – 115200 Kbps (SW Selectable)
- Data Bits 8
- Parity None
- CRC Yes
- Slave ID 1, DIP (SW Selectable)
- Function Code 0X03 (Read Holding Register)

## Recommended Cable Electrical Characteristics

|                                          |                                           |
|------------------------------------------|-------------------------------------------|
| <b>22 AWG Cable</b>                      | Shielded and twisted pair should be used. |
| <b>Tinned Copper</b>                     | Recommended                               |
| <b>Nominal Conductor DCR</b>             | 14.7 ohm / 1000 ft                        |
| <b>Nominal Capacitance</b>               | 11 pf / feet (conductor to conductor)     |
| <b>High Frequency Non-Insertion Loss</b> | 0.5db / 100ft                             |

## 5. Modbus RS485 Data Storage register

| ID | Function Description                                   | Register Description | Modbus Function Code | Protocol | Data Type  |
|----|--------------------------------------------------------|----------------------|----------------------|----------|------------|
| 1  | Temperature                                            | 40001                | 0x03                 | RS485    | 16-bit int |
|    |                                                        |                      |                      |          |            |
| 1  | Display Baud Rate (Default: 960)                       | 40011                | 0x03                 | RS485    | 16-bit int |
| 2  | Enter New Baud Rate                                    | 40012                | 0x03                 | RS485    | 16-bit int |
| 3  | Display Slave ID (Default: 1)                          | 40013                | 0x03                 | RS485    | 16-bit int |
| 4  | Enter New Slave ID                                     | 40014                | 0x03                 | RS485    | 16-bit int |
| 5  | Display Parity (Default: None-3 And for Odd-1, Even-2) | 40015                | 0x03                 | RS485    | 16-bit int |
| 6  | Enter New Parity                                       | 40016                | 0x03                 | RS485    | 16-bit int |
| 7  | Display Stop Bit (start-2/stop bit-1)                  | 40017                | 0x03                 | RS485    | 16-bit int |
| 8  | Enter New Start/ Stop Bit                              | 40018                | 0x03                 | RS485    | 16-bit int |

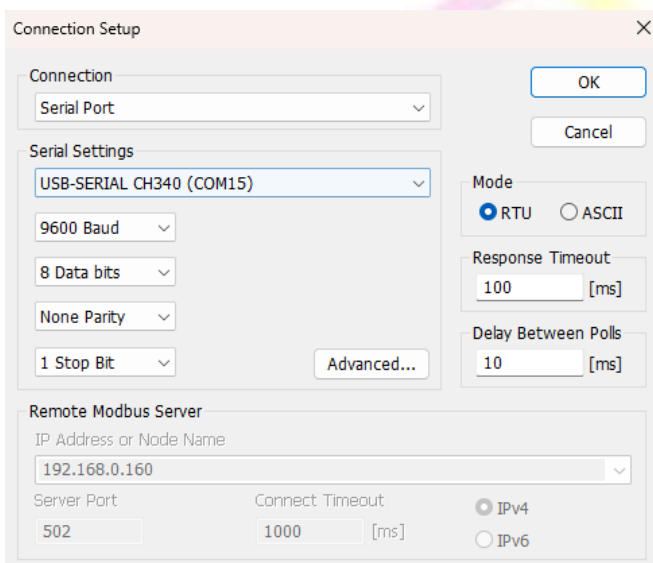
## 6. Modbus Configuration

Here is an example of receiving data on Mb poll software.

- Open Mb-poll software connects using configuration setting (Default baud rate is 9600 and slave id is 1).
- After connecting you will get data as shown below.

### Important Note

1. If you want to enter the baud rate 9600/19200/38400/57600/115200etc, enter it like this 960/1920/3840/5760/11520etc.
2. After that press the reset button or reboot the system.
3. Then you will get the present value of 11520 at 40011 baud rate (115200) and 2 at 40013 slave id.



**Connection Setup**

Connection: Serial Port

Serial Settings: USB-SERIAL CH340 (COM15)

Mode: ☒ RTU ☐ ASCII

Response Timeout: 100 [ms]

Delay Between Polls: 10 [ms]

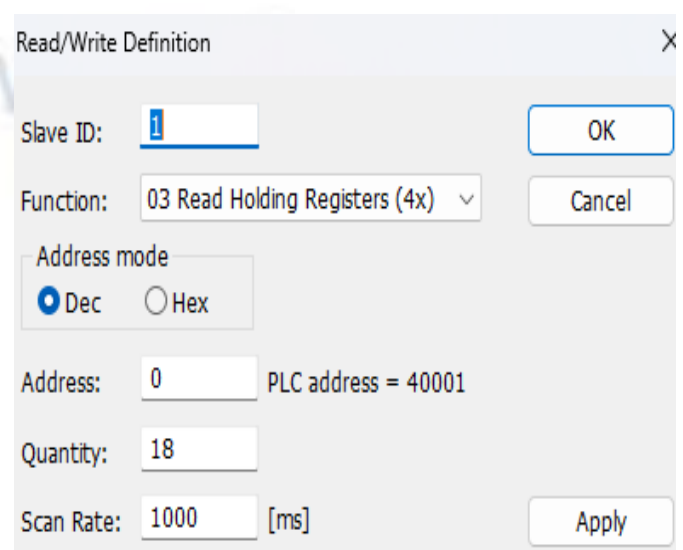
Remote Modbus Server

IP Address or Node Name: 192.168.0.160

Server Port: 502

Connect Timeout: 1000 [ms]

☒ IPv4 ☐ IPv6



**Read/Write Definition**

Slave ID: 1

Function: 03 Read Holding Registers (4x)

Address mode: ☒ Dec ☐ Hex

Address: 0 PLC address = 40001

Quantity: 18

Scan Rate: 1000 [ms]

Mbpoll1

Tx = 2844: Err = 51: ID = 1: F = 03: SR = 1000ms

|   | Alias       | 00000 | Alias           | 00010 |
|---|-------------|-------|-----------------|-------|
| 0 | Temperature | 320   | Current Baud    | 960   |
| 1 |             | 0     | New Baud        | 0     |
| 2 |             | 0     | Current SID     | 1     |
| 3 |             | 0     | New SID         | 0     |
| 4 |             | 0     | Current Parity  | 3     |
| 5 |             | 0     | New Parity      | 0     |
| 6 |             | 0     | Current Stopbit | 1     |
| 7 |             | 0     | New Stopbit     | 0     |
| 8 |             | 0     |                 | 0     |
| 9 |             | 0     |                 | 0     |

- Above image Shows sensor data over Modbus with default Modbus settings
- At register 40001 shows Temperature Values.
- The value 320 indicates 32.0 temperature.
- As you can see in the image above, the resistor 40011 – 960 represents the present bund rate (9600) of the board and 40013 – 1 represents the present slave id of the board.
- To change the baud rate and slave id Jumper should be in single/removed then, you must enter a value on the 40012 resistors for the baud rate and 40014 for the slave id as shown in the image below and restart the board.
- Now you can see data below image set 115200 baud rate and slave ID 10.

Mbpoll1

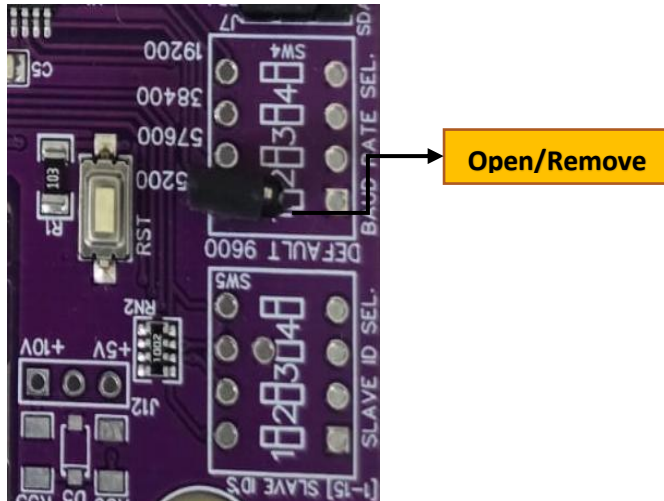
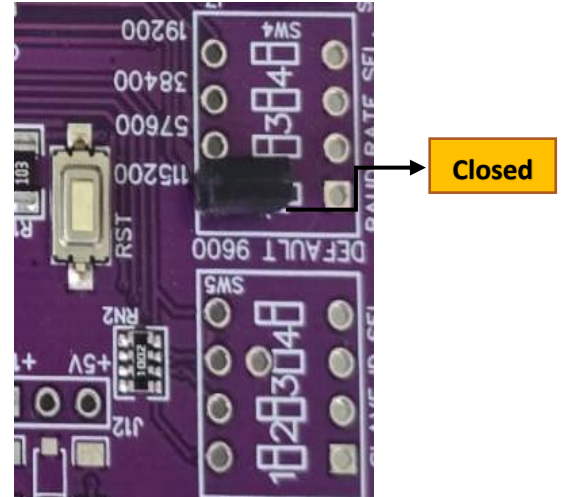
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|---|-------------|-------|-----------------|-------|
| 0 | Temperature | 320   | Current Baud    | 960   |
| 1 |             | 0     | New Baud        | 11520 |
| 2 |             | 0     | Current SID     | 1     |
| 3 |             | 0     | New SID         | 10    |
| 4 |             | 0     | Current Parity  | 3     |
| 5 |             | 0     | New Parity      | 0     |
| 6 |             | 0     | Current Stopbit | 1     |
| 7 |             | 0     | New Stopbit     | 0     |
| 8 |             | 0     |                 | 0     |
| 9 |             | 0     |                 | 0     |

Mbpoll1

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|---|-------------|-------|-----------------|-------|
| 0 | Temperature | 320   | Current Baud    | 11520 |
| 1 |             | 0     | New Baud        | 0     |
| 2 |             | 0     | Current SID     | 10    |
| 3 |             | 0     | New SID         | 0     |
| 4 |             | 0     | Current Parity  | 3     |
| 5 |             | 0     | New Parity      | 0     |
| 6 |             | 0     | Current Stopbit | 1     |
| 7 |             | 0     | New Stopbit     | 0     |
| 8 |             | 0     |                 | 0     |
| 9 |             | 0     |                 | 0     |

**Image1.1****Image1.2**

### **Hard reset Setting**

- Somehow you forgot the baud rate or slave id of the board then Place the jumper (as shown in the image 1.2) from the board reboot the system or press the reset button (given on the board). Then you will get the default baud rate of 9600 and slave id 1.

### **Contact Information**

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